

Ex. No: 7

Use AFLogical OSE to extract data from an Android

Date:

Description:

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

1. Download AFLogical OSE:

- Visit the AFLogical OSE GitHub page and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java Make sure you have Java installed on your machine. You can download it from the official Java website.

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the Android Developer's website and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings About Phone and tap Build Number times to enable Developer Options.
- Go to Settings Developer Options and enable USB Debugging

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:
adb devices

Download AFLogical OSE:

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE: Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device: `bash adb install aflogical.apk`
- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- You'll see options to extract various types of data (eg, contacts, SMS, MMS, call logs). Select the data types you to extract
- On the Android device, open the AFLogical app.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer.
Run:
`bash adb pull/sdcard/aflogical/path/to/destination`
- Replace/path/to/destination with the directory path where you want to save the extracted data on your computer.

1. Verify the Data:

Navigate to the destination directory and check the esv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running.

Bash adb

uninstallcom.viaforensics.android.aflogical

2. Disconnect the Device:

- Safely disconnect the Android device from your computer. By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Output:

Installing JAVA:

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sarvegna>java -version
java version "26.0.2.1" 2026-08-18
Java(TM) SE Runtime Environment (build 26.0.2.1+1-7)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.2.1+1-7, mixed mode, sharing)

C:\Users\Sarvegna>
```

Installing ADB Tools:

```
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

ADB device connection:

```
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
ZD2227X3N4      device

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

Installing aflogical.apk:

```
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>adb install
com.viaforensics.android.aflogical_ose_1.5.2_QSE_152.apk
Performing Streamed Install
Success

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

Capturing SMS and Call logs:

```
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe shell content query --uri
content://sms > sms.txt

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe shell content query --uri
content://call_log/calls > call_logs.txt

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

Extracted SMS logs:

Row: 0_id=6806, thread_id=2294, address=9f19a2675638, person=NULL, date=1787599138655, date_sent=1787599138658, protocol=0, read=0, status=1, type=1, reply_path_present=0, subject-proto(cjzkMwb5Snb2Bn8GdUWSkczRpXCShWdRLlLlcZwW22uLy4SFClAhkDSwRt8s3QNR7plnz57E6PK, body=Bear Customer, You have a missed call from +9194932675638. The last missed call was at 16:58 AM on 07-Sep-2026. Thankyou, Team Jio, service_center=+919044900412, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1

Row: 1_id=6807, thread_id=1887, address=3KIHSE5WS-5, person=NULL, date=1788844789072, date_sent=1788844785600, protocol=0, read=0, status=1, type=1, reply_path_present=0, subject-proto(cjzkMwb5Snb2Bn8GdUWSkczRpXCShWdRLlLlcZwW22uLy4SFClAhkDSwRt8s3QNR7plnz57E6PK, body=GNDMAINVESTECOFHRRIVATEJTED, on 29-08-2026 reported your F&Kmail, Rs.8,000 0 securities balwith The broker-NSE, service_center=+917388307506, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1

Prateek id=gddg id=mkks AH083 messaging, seen=1

Row: 2_id=6808, thread_id=2237, address=3X-TSRSCS-5, person=NULL, date=1788533923394, date_sent=1788533919800, protocol=0, read=0, status=1, type=1, reply_path_present=0, subject-proto(cjzkMwb5Snb2Bn8GdUWSkczRpXCShWdRLlLlcZwW22uLy4SFClAhkDSwRt8s3QNR7plnz57E6PB, body=RIR444442B174

Tn:422715

Dt:-05-09-26 Dep.Time:-09:47 Hrs.

Cms KRNT to NDU

Flgs:SLL

P1-94,34

P2-54,29

P3-54,32

Boarding allowed from KRNT only

Chart Prepared

Url for coach position: <https://enquiry.indianrail.gov.in/wntes/c/wag6fjNwa9d7WejngKd81>

Download Railtone for latest updates <https://railtone.indianrailways.gov.in>

For Enquiry/Complaint/Assistance, please dial 139 FR-CRIS, service_center=+91701978001, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1

Row: 3_id=6807, thread_id=3771, address=3KIHSE5WS-5, person=NULL, date=178853367713, date_sent=1788533635000, protocol=0, read=0, status=1, type=1, reply_path_present=0, subject-proto(cjzkMwb5Snb2Bn8GdUWSkczRpXCShWdRLlLlcZwW22uLy4SFClAhkDSwRt8s3QNR7plnz57E6PK, body=Bearer white details based on unsolicited trip through Whatsapp, telegram,SMS,calls,YouTube,etc& take an informed decision. Visit <https://bit.ly/3QXSKSO>-NSE, service_center=+9170204705016, locked=0, sub_id=2, error_code=1, creator=com.google.android.apps.messaging, seen=1

Row: 4_id=6806, thread_id=1886, address=3JJ1P6F4Y-5, person=NULL, date=1788532742702, date_sent=1788532691800, protocol=0, read=0, status=1, type=1, reply_path_present=0, subject-proto(cjzkMwb5Snb2Bn8GdUWSkczRpXCShWdRLlLlcZwW22uLy4SFClAhkDSwRt8s3QNR7pmaIna75Spk, body="");

Auto 04-Sep-26 17:21 వీడువచ్చింది రిడైరెక్ట్ చేయండి 100% విషయం.

పేరు నమోదు: 8874999510

లేదానా లేదా రికార్డు: 2 GB

లేదానా లేదా క్లౌడ్: 09:27 వచ్చి అందుబాటులో ఉంటుంది.

మీ వద్ద అందువల్ల లేదా వ్యాక్ రికార్డుతో, మీ ఇంటర్నెట్ వేగం 66 kbps వరకు తగ్గించండి.

M/Jio యూజర్ ద్వారా లేదా వ్యాక్ రికార్డు వేగం: <http://tiny.jio.com/dhyudatapack>

Extracted Call logs:

```

Row: 0 formatted_number=NULL, duration=50, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=NULL, features=68, transcription=NULL, last_modified=1729788812435, _id=8385, new=0, date=1718892698184, name=Seenu, type=1, presentation=1, via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1,
Row: 1 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=NULL, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=NULL, features=1, transcription=NULL, last_modified=1729788812435, _id=8386, new=0, date=1718892698184, name=Seenu, type=1, presentation=1, via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1,
Row: 2 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=NULL, features=1, transcription=NULL, last_modified=1729788812435, _id=8386, new=0, date=1718892698184, name=Seenu, type=1, presentation=1, via_number=, numberlabel=Mobile, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, video_call=0,
Row: 3 formatted_number=+91 77991 68016, duration=38, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number=+917799168016, countryiso=IN, cnap_name=+917799168016, photo_uri=, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=0, features=64, transcription=NULL, last_modified=1729788812435, _id=8499, new=0, date=1720318990704, name=, type=1, presentation=1, via_number=, numberlabel=Mobile, normalized_number=+917799168016,

```

Extracted Device Properties:

```
PS C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Device Model:" (.\adb.exe shell
getprop ro.product.model)
Device Model: moto g62 5G
PS C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Android Version:" (.\adb.exe
shell getprop ro.build.version.release)
Android Version: 13
PS C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Security Patch Level:" (
.\adb.exe shell getprop ro.build.version.security_patch)
Security Patch Level: 2025-07-01
PS C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

Uninstalling aflogical.apk:

```
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>.\adb.exe uninstall --user 0  
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk  
  
C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>
```

Result:

The experiment successfully performed a logical extraction of data from the Android device using the AF Logical OSE framework via ADB interface connection. The tool bypassed file system access limitations to cleanly pull device properties, SMS logs, and call records into structured.csv files, which were verified on the local host before safely uninstalling the APK tool.

